

SS

BSDS MATHEMATICS I MIDTERM EXAMINATION

Date: 7th October, 2025

Time: 11 am to 1 pm

Attempt all questions.

Total Marks: 40

- (1) For each $n \in \mathbb{N}$, consider the finite set $S_n = \{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}\}$. Hence $S_1 = \{1\}$, $S_2 = \{1, \frac{1}{2}\}$, $S_3 = \{1, \frac{1}{2}, \frac{1}{3}\}$ and so on. Consider the sequence (y_n) obtained by writing the elements of all the S_n 's one after another, starting with S_1 , then the elements of S_2 , then the elements of S_3 , and so on. So the sequence (y_n) is $(1, 1, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{3}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 1, \dots)$. Calculate $\liminf y_n$ and $\limsup y_n$. First clearly define $\liminf$ and $\limsup$ of a sequence of real numbers. (5 + 5 = 10 marks)
- (2) Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = 0$ if x is an irrational real number, $f(x) = \frac{1}{q}$ if $x = \frac{p}{q}$ is a nonzero rational number and p and q have no common factors, and $f(0) = 1$. Prove that f is discontinuous at all rational numbers. First, clearly write what it means to show that a real valued function f is discontinuous at a point a in the domain. (5 + 5 = 10 marks)
- (3) Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = x^2$ for x is rational, and $f(x) = 0$ for x irrational. What is the set S of points of $\mathbb{R}$ at which f is continuous and what is the set T of points of $\mathbb{R}$ at which f is differentiable? Write clearly what it means for a real valued function f to be differentiable at a point a of the domain. (5 + 5 = 10 marks)
- (4) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \cos(e^{(x^2-3x)})$. Find the derivative of f at every point x using the chain rule. State the chain rule for derivatives clearly and show how you are using this theorem to do the computation. You may assume that the derivatives of $\cos(x)$ and e^x are $-\sin(x)$ and e^x , respectively. (5 + 5 = 10 marks)